Navid Bahadoran

Problem Set I

1. Derive the autocorrelation function of

$$y_t = 1 + \varepsilon_t + 0.6\varepsilon_{t-1} + 0.2\varepsilon_{t-2},$$

where $\varepsilon_t \sim WN(0,1)$, i.e., a white noise process with zero mean and unit variance. Show your work.

As we know, white noise samples are uncorrelated.

$Var(y_t) = Var(y_t) + Var(\epsilon_t) + 0.36Var(\epsilon_{t-1}) = 0.04Var(\epsilon_{t-2})$

$\Rightarrow \gamma_0 = 1 + 0.36 + 0.04 = 1.4$

$\gamma_1 = Cov(y_t, y_{t-1}) = \mathbb{E}[y_t y_{t-1}] - \mathbb{E}[y_t]\mathbb{E}[y_{t-2}] =$
$= \mathbb{E}([1 + \epsilon_t + 0.6\epsilon_{t-1} + 0.2\epsilon_{t-2}][1 + \epsilon_{t-1} + 0.6\epsilon_{t-2} + 0.2\epsilon_{t-3}])$
$- \underbrace{\mathbb{E}[1 + \epsilon_t + 0.6\epsilon_{t-1} + 0.2\epsilon_{t-2}]}_{1} \underbrace{\mathbb{E}[1 + \epsilon_{t-1} + 0.6\epsilon_{t-2} + 0.2\epsilon_{t-3}]}_{1}$
$= 1 + 0.6 + 0.12 - 1 = 0.72$

$\gamma_2 = Cov(y_t, y_{t-2}) = \mathbb{E}[y_t y_{t-2}] - \mathbb{E}[y_t]\mathbb{E}[y_{t-2}] =$
$= \mathbb{E}([1 + \epsilon_t + 0.6\epsilon_{t-1} + 0.2\epsilon_{t-2}][1 + \epsilon_{t-2} + 0.6\epsilon_{t-3} + 0.2\epsilon_{t-4}])$
$- \underbrace{\mathbb{E}[1 + \epsilon_t + 0.6\epsilon_{t-1} + 0.2\epsilon_{t-2}]}_{1} \underbrace{\mathbb{E}[1 + \epsilon_{t-2} + 0.6\epsilon_{t-3} + 0.2\epsilon_{t-4}]}_{1}$
$= 1 + 0.2 - 1 = 0.2$

$\gamma_3 = Cov(y_t, y_{t-3}) = \mathbb{E}[y_t y_{t-3}] - \mathbb{E}[y_t]\mathbb{E}[y_{t-3}] =$
$= \mathbb{E}([1 + \epsilon_t + 0.6\epsilon_{t-1} + 0.2\epsilon_{t-2}][1 + \epsilon_{t-3} + 0.6\epsilon_{t-4} + 0.2\epsilon_{t-5}])$
$- \underbrace{\mathbb{E}[1 + \epsilon_t + 0.6\epsilon_{t-1} + 0.2\epsilon_{t-2}]}_{1} \underbrace{\mathbb{E}[1 + \epsilon_{t-3} + 0.6\epsilon_{t-4} + 0.2\epsilon_{t-5}]}_{1} = 1 - 1 = 0$

$\gamma_h = 0 \; for \; h \geq 3$

$\rho_0 = \frac{\gamma_0}{\gamma_0} = 1, \rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{0.72}{1.4} = 0.514, \rho_2 = \frac{\gamma_2}{\gamma_0} = \frac{0.2}{1.4} = 0.143$
$\rho_h = 0 \; for \; h \geq 3$

2. Consider the time series

$$y_t = \beta_0 + \beta_1 t + \varepsilon_t + \theta\varepsilon_{t-1},$$

where $\varepsilon_t \sim WN\left(0, \sigma^2\right)$, and β_0, β_1, and θ are fixed constants ($\beta_1 \neq 0$).

(a) Determine whether y_t is weakly stationary.

(b) Determine whether $x_t = y_t - y_{t-1}$ is weakly stationary.

a)

For the process to be weakly stationary, it must have constant mean and variance. Let's examine the mean first:

$\mathbb{E}[y_t] = \mathbb{E}[\beta_0 + \beta_1 t + \epsilon_t + \theta\epsilon_{t-1}] = \beta_0 + \beta_1 t$ which is not constant and has time dependency. **So, y_t is not weakly stationary.**

b)

$x_t = y_t - y_{t-1} = \beta_0 + \beta_1 t + \epsilon_t + \theta\epsilon_{t-1} - [\beta_0 + \beta_1(t-1) + \epsilon_{t-1} + \theta\epsilon_{t-2}]$
$= \beta_1 + \epsilon_t + (\theta - 1)\epsilon_{t-1} - \theta\epsilon_{t-2}$ which is MA(2), and as we know MA processes are always weakly stationary. **So, x_t is weakly is stationary.**

3. Consider the $MA(1)$ process:

$$y_t = \varepsilon_t + \theta\varepsilon_{t-1}, \quad \varepsilon_t \sim WN\left(0, \sigma^2\right).$$

(a) Suppose $|\theta| < 1$. Show that ε_t can be expressed in terms of y_s, $s \leq t$. Do more recent observations have higher weights than observations from the distant past?

(b) Suppose now $|\theta| > 1$. Show that ε_t can be expressed in terms of y_s, $s \geq t$.
Hint: Instead of doing backward substitution, use forward substitution starting from $\varepsilon_t = \theta^{-1}\varepsilon_{t+1} + \theta^{-1}y_{t+1}$.
Comment: Note that in this case the series you obtained in (a) does not converge, and it indicates that more distant observations have a greater influence on the current error.

a)

$y_t = \epsilon_t + \theta\epsilon_{t-1}$ (1)

$and\ y_{t-1} = \epsilon_{t-1} + \theta\epsilon_{t-2} \Rightarrow \theta\epsilon_{t-1} = \theta y_{t-1} - \theta^2\epsilon_{t-2}$ (2)

$\xrightarrow{from\ (1) and\ (2)} y_t = \epsilon_t + \theta y_{t-1} - \theta^2\epsilon_{t-2} \Rightarrow \epsilon_t = y_t - \theta y_{t-1} + \theta^2\epsilon_{t-2}$

By recursive reasoning we have:

$\epsilon_t = \sum_{i=0}^{k}(-\theta)^i y_{t-i} + (-\theta)^{k+1}\epsilon_{t-(k+1)},\ k \geq 0$

By taking the limit to infinity

$$\Rightarrow \epsilon_t = \lim_{k\to\infty}\left[\sum_{i=0}^{k}(-\theta)^i y_{t-i} + (-\theta)^{k+1}\epsilon_{t-(k+1)}\right] = \sum_{i=0}^{\infty}(-\theta)^i y_{t-i}$$

b)

$$y_t = \epsilon_t + \theta\epsilon_{t-1} \Rightarrow y_{t+1} = \epsilon_{t+1} + \theta\epsilon_t \Rightarrow \epsilon_t = \theta^{-1}y_{t+1} - \theta^{-1}\epsilon_{t+1}$$

By recursive reasoning we have:

$$\epsilon_t = \sum_{i=1}^{k} -(-\theta)^{-i} y_{t+i} + (-\theta)^{-k}\epsilon_{t+k},\ k \geq 1$$

$$\Rightarrow \epsilon_t = \lim_{k\to\infty}\left[\sum_{i=1}^{k} -(-\theta)^{-i} y_{t+i} + (-\theta)^{-k}\epsilon_{t+k}\right] = \sum_{i=1}^{\infty} -(-\theta)^{-i} y_{t+i}$$

4. Let

$$x_t = \varepsilon_t - 0.4\varepsilon_{t-1}, \quad \varepsilon_t \sim WN(0,4),$$
$$y_t = u_t - 0.7u_{t-1}, \quad u_t \sim WN(0,1),$$

where $\{\varepsilon_t\}$ is distributed independently of $\{u_t\}$. Consider $z_t = x_t + y_t$.

(a) Find the autocovariances of $\{z_t\}$.

(b) Express z_t as an invertible MA process.
Hint: Suppose that $z_t = e_t + \theta e_{t-1}$, where $e_t \sim WN(0, \sigma_e^2)$. You will need to solve for θ and σ_e^2 such that z_t has the same autocovariances obtained in part (a).

a) Since the ϵ_t and u_t are independent, autocovariance z_t is the summation of autocovariance x_t and y_t. Moreover, x_t and y_t are both MA(1) and considering that MA has a zero mean, we have:

$\gamma_{x,0} = \mathbb{E}[x_t^2] = 4 + 0.16 \times 4 = 4.64$
$\gamma_{x,1} = \mathbb{E}[x_t x_{t-1}] = (-0.4) \times 4 = -1.6, \qquad \gamma_{x,h} = 0, h > 1$
$\gamma_{y,0} = \mathbb{E}[y_t^2] = 1 + 0.49 \times 1 = 1.49$
$\gamma_{y,1} = \mathbb{E}[y_t y_{t-1}] = (-0.7) \times 1 = -0.7, \qquad \gamma_{y,h} = 0, h > 1$

$\Rightarrow \gamma_{z,0} = \gamma_{x,0} + \gamma_{y,0} = 4.64 + 1.49 = 6.13$
And $\gamma_{z,1} = \gamma_{x,1} + \gamma_{y,1} = -1.6 - 0.7 = -2.3$, $\gamma_{z,h} = 0, h > 1$

b)
$z_t = e_t + \theta e_{t-1} \Rightarrow \gamma_{z,0} = (1+\theta^2)\sigma_e^2$ *and* $\gamma_{z,1} = \theta\sigma_e^2$
From part (a) we have:

$\gamma_{z,0} = 6.13 \Rightarrow (1+\theta^2)\sigma_e^2 = 6.13$ (1)

,

$\gamma_{z,1} = -2.3 \Rightarrow \theta\sigma_e^2 = -2.3$ (2)

$\xrightarrow{from\ (1)\ and\ (2)} \theta = -0.452$. The other root is not acceptable as we are looking for invertible MA process.

$\sigma_e^2 = \dfrac{2.3}{0.452} = 5.088$

$\Rightarrow z_t = e_t - 0.452e_{t-1}, \qquad e_t \sim N(0, 5.088)$

5. Consider the following $AR(2)$ process:

$$y_t = y_{t-1} - 0.24y_{t-2} + \varepsilon_t, \quad \varepsilon_t \sim WN\left(0, \sigma^2\right).$$

(a) Express this process using the lag operator L.

(b) Is this process weakly stationary? Does y_t exhibit stochastic business cycles?

(c) Find the coefficients $\{\psi_i\}$ in the MA representation $y_t = \sum_{i=0}^{\infty} \psi_i \varepsilon_{t-i}$.

(d) Consider now

$$x_t = 2 + x_{t-1} - 0.24x_{t-2} + \varepsilon_t.$$

Find the MA representation of x_t. (*Hint*: First find the mean of x_t and rewrite this process as an $AR(2)$ with zero mean.)

(e) Check for stationarity and stochastic business cycles for the process

$$z_t = z_{t-1} - 0.3z_{t-2} + \varepsilon_t.$$

Find the average length of the stochastic cycles.

a)

$(1 - L + 0.24L^2)y_t = \epsilon_t$

$b)$

$1 - z + 0.24z^2 = 0 \Rightarrow z_{1,2} = \frac{5}{3}, \frac{5}{2}$. Since the roots are greater than one, the process is stationary. But since none of the roots are complex number, the process does not exhibit stochastic cycles.

c)

$$y_t = (1 - L + 0.24L^2)^{-1}\epsilon_t = \frac{3}{1 - 0.6L}\epsilon_t - \frac{2}{1 - 0.4L}\epsilon_t$$

$$y_t = 3\sum_{i=0}^{\infty} 0.6^i \epsilon_{t-i} - 2\sum_{i=0}^{\infty} 0.4^i \epsilon_{t-i} = \sum_{i=0}^{\infty} \left(3 \times 0.6^i - 2 \times 0.4^i\right)\epsilon_{t-i}$$

$\Rightarrow \psi_i = 3 \times 0.6^i - 2 \times 0.4^i$

d)

From Part (c) we have:

$$(1 - L + 0.24L^2)x_t = 2 + \epsilon_t \Rightarrow x_t = 2(1 - L + 0.24L^2)^{-1} + (1 - L + 0.24L^2)^{-1}\epsilon_t$$

$$\Rightarrow x_t = \frac{2}{1 - 1 + 0.24} + \sum_{i=0}^{\infty}\left(3 \times 0.6^i - 2 \times 0.4^i\right)\epsilon_{t-i} = 8.33 + \sum_{i=0}^{\infty}\left(3 \times 0.6^i - 2 \times 0.4^i\right)\epsilon_{t-i}$$

$$\Rightarrow x_t - 8.33 = \sum_{i=0}^{\infty}\left(3 \times 0.6^i - 2 \times 0.4^i\right)\epsilon_{t-i}$$

e)

$$(1 - L + 0.3L^2)z_t = \epsilon_t \Rightarrow 1 - z + 0.3z^2 = 0 \Rightarrow z_{1,2} = 1.667 \pm 0.745356i$$

Since the modules of roots are greater than one, the process is stationary. Also, because of all roots are complex the process exhibits business cycles.

$$k = \frac{2\pi}{\theta} = \frac{2\pi}{\cos^{-1}\left(\frac{\phi_1}{2\sqrt{-\phi_2}}\right)} = \frac{2\pi}{\cos^{-1}(1/2\sqrt{0.3})} = 14.94$$